



Teleperm XS Implementation & Upgrading of Safety I&C

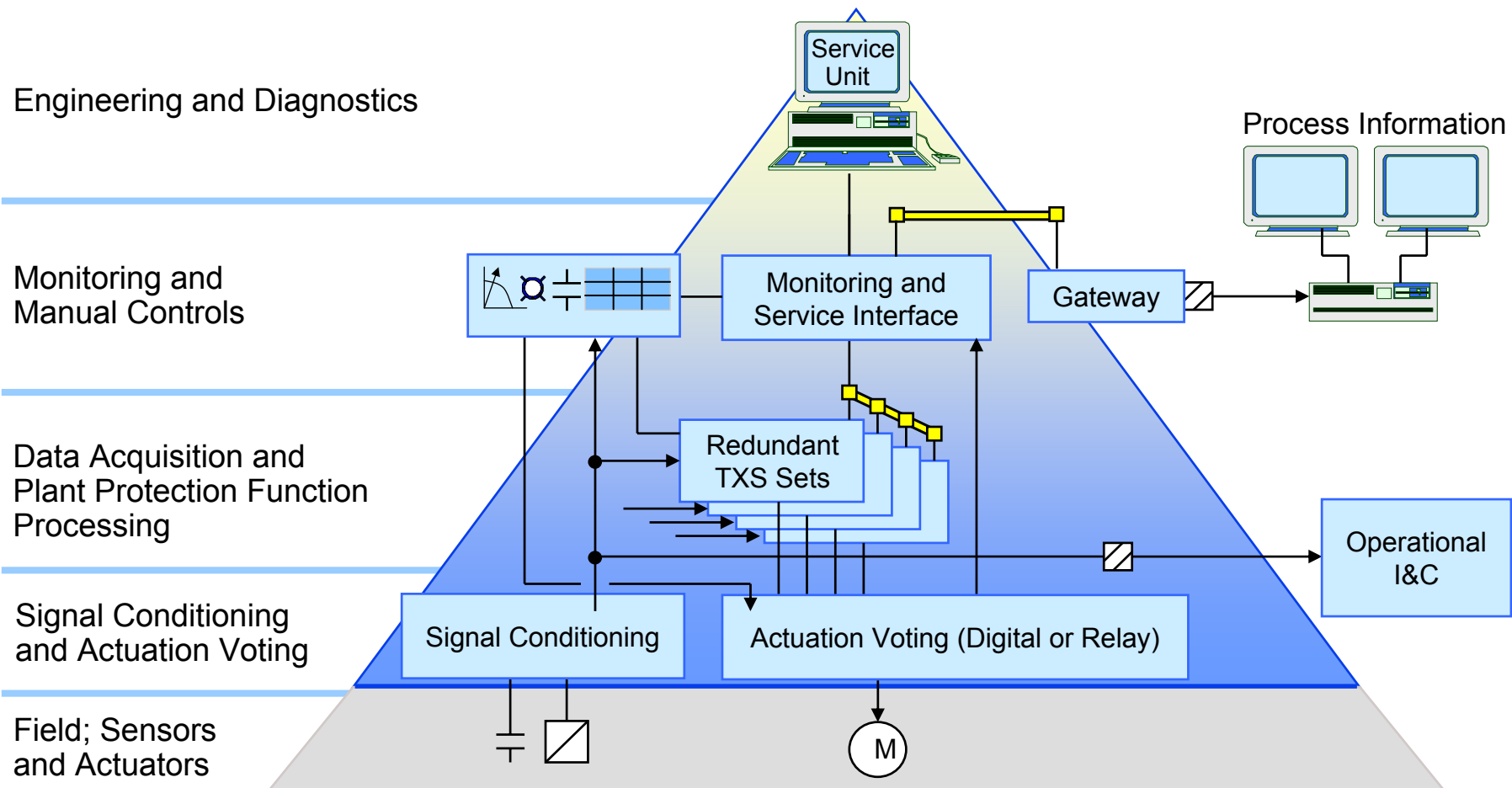
Günter Niedzballa

Framatome ANP

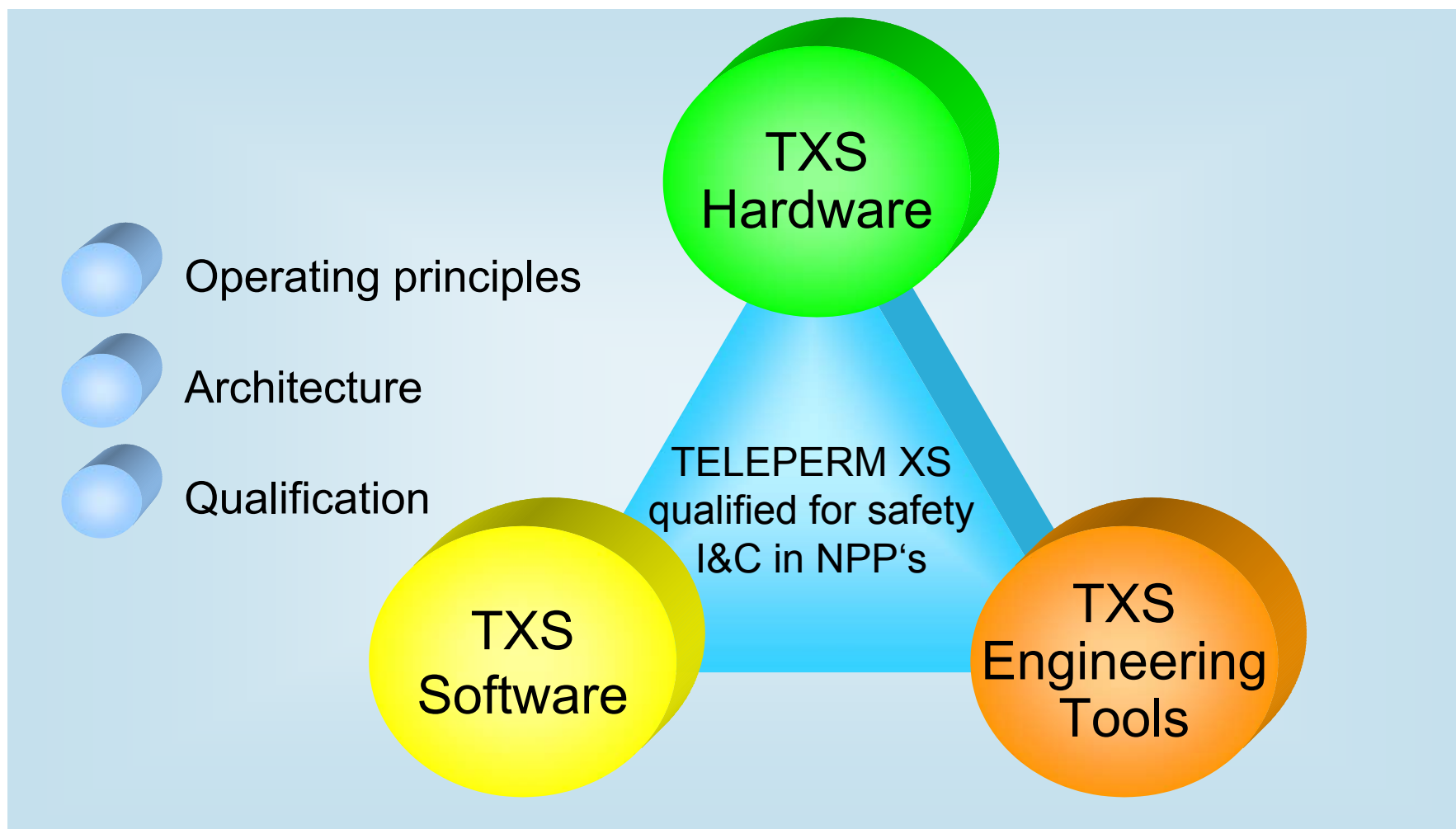
System Platform, Structures, Qualification, Engineering Process,
Tests, References, Operational Experience, Outlook

TXS System Platform

TELEPERM XS Processing Levels



TELEPERM XS System Platform

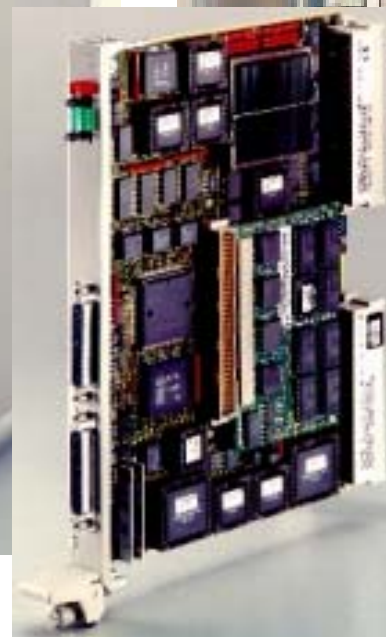
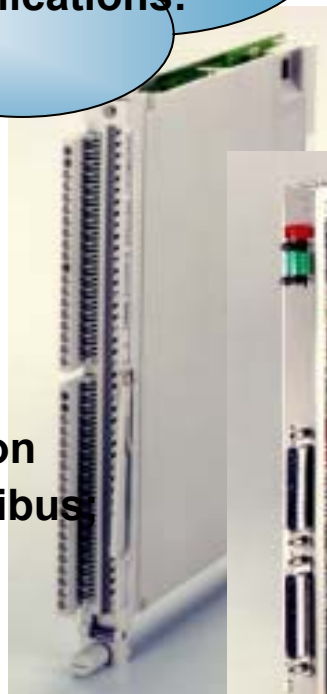


TELEPERM XS Hardware: Components

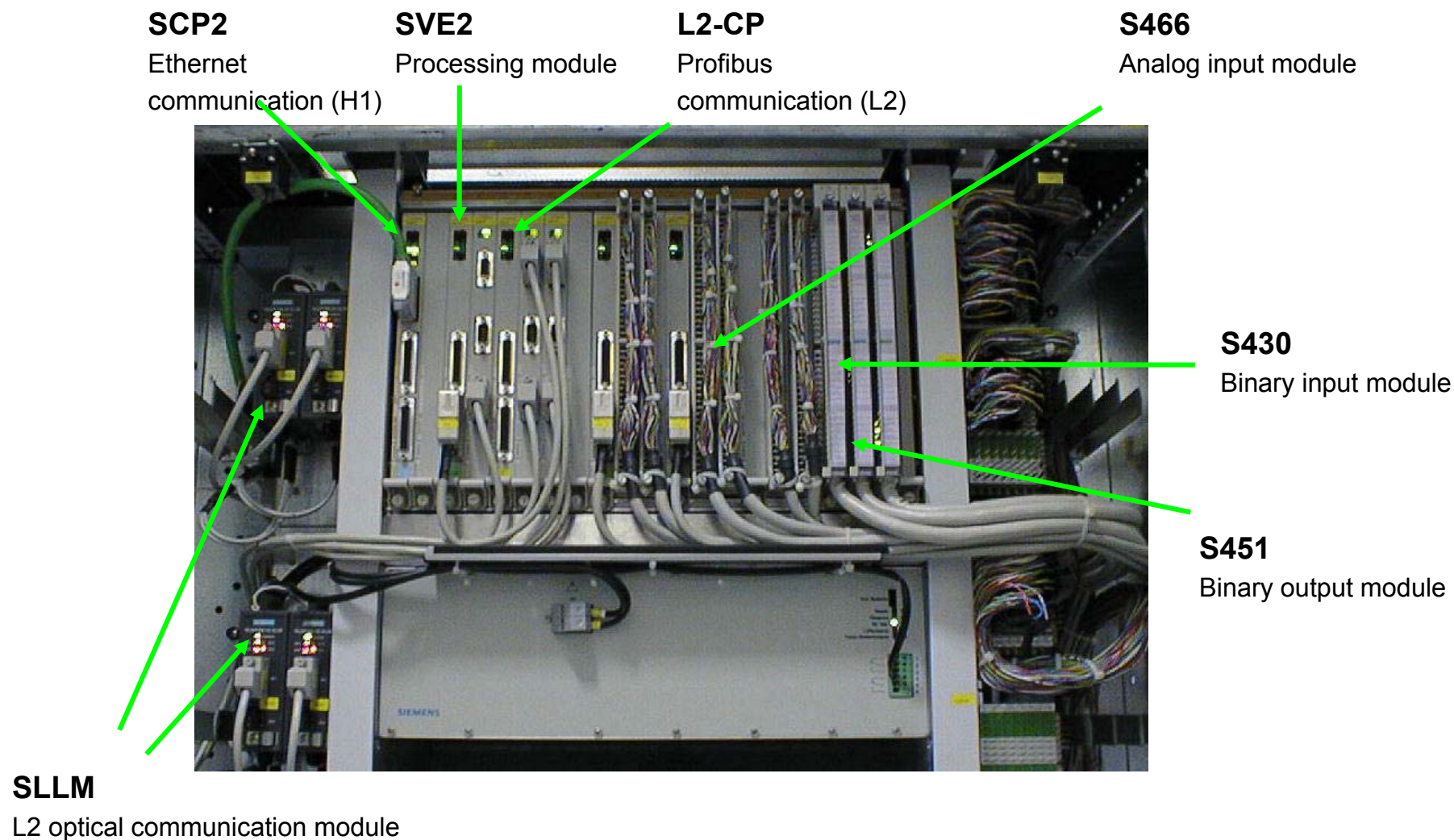
**Standard hardware based on
SIMATIC, SIMICRO, SINEC**

- broad industrial operating experience
- qualified for safety grade applications:
type-tested

- Cabinet
- Sub-Racks
- Processing and communication components (Ethernet or Profibus, electrical or fibre-optical)
- I/O modules

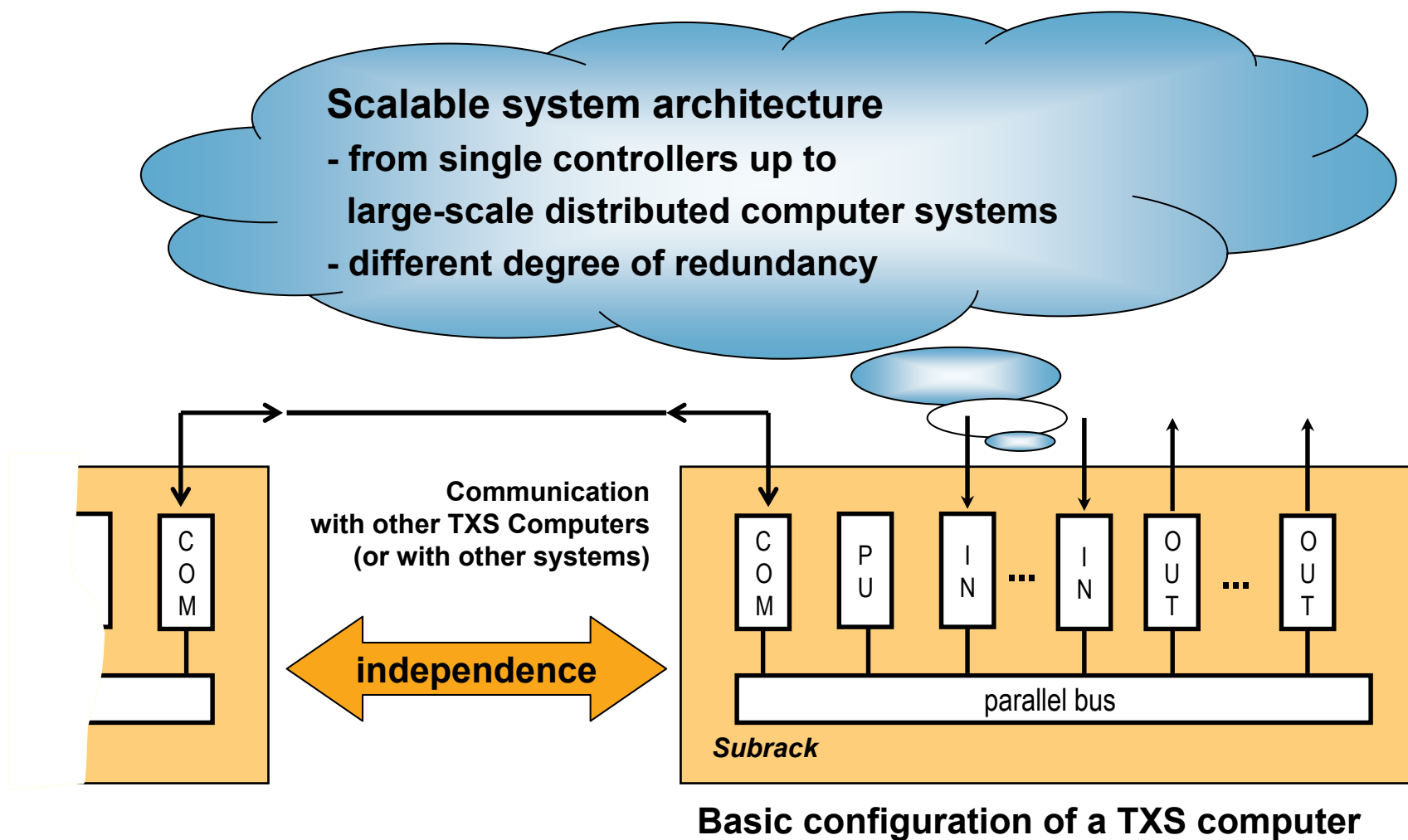


TELEPERM XS Hardware: Example Configuration

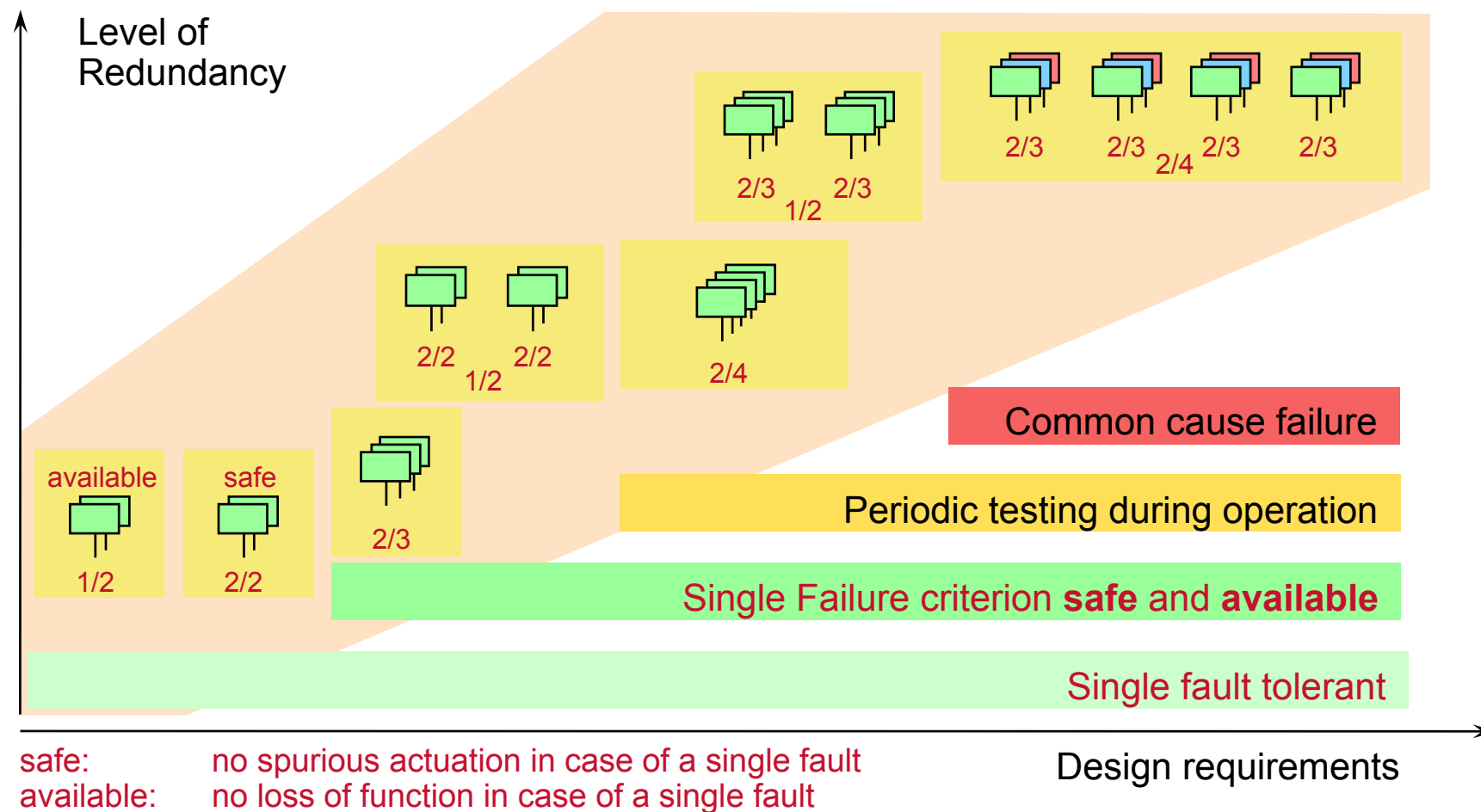


TXS Structures

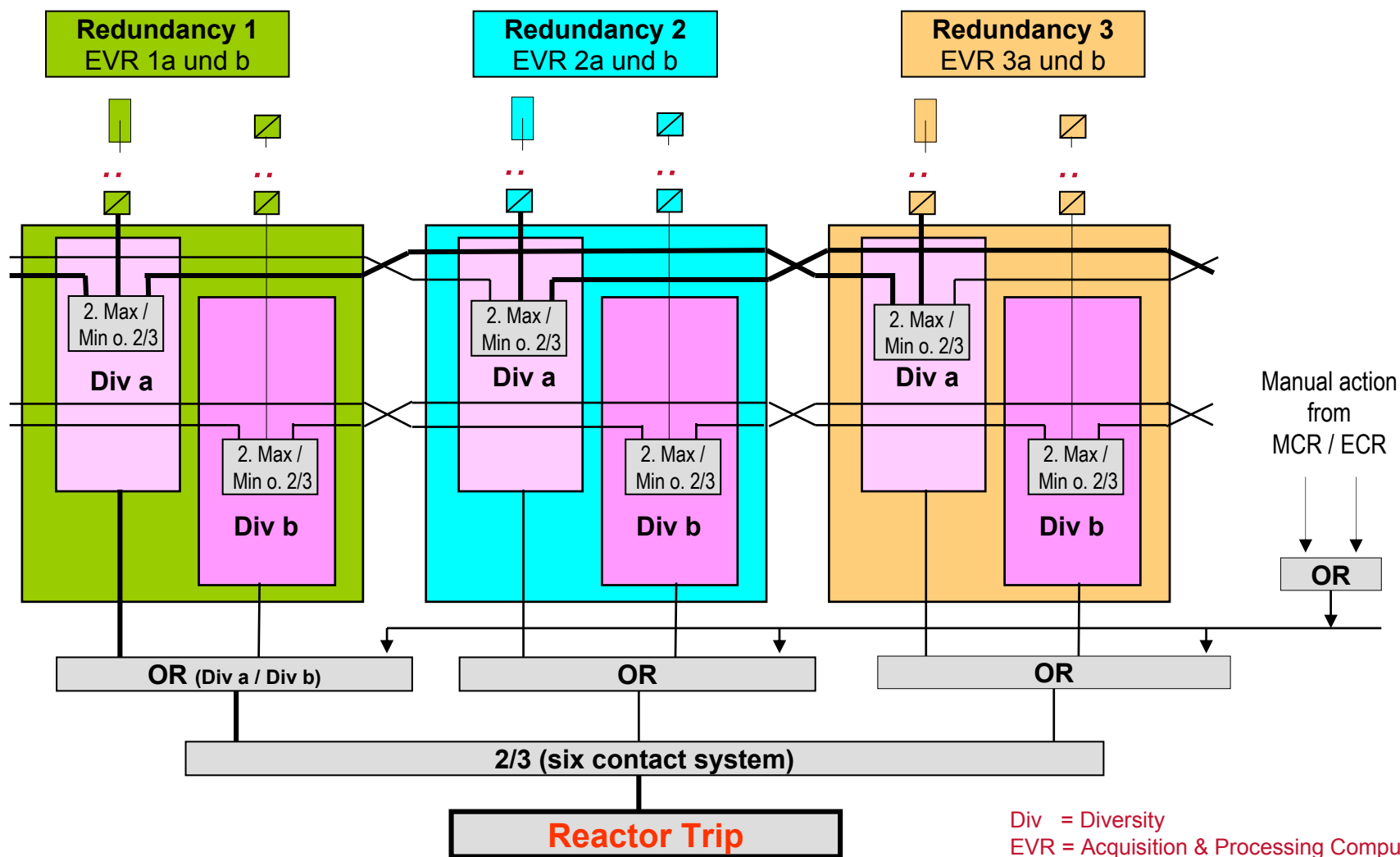
TELEPERM XS Hardware: Configuration Flexibility



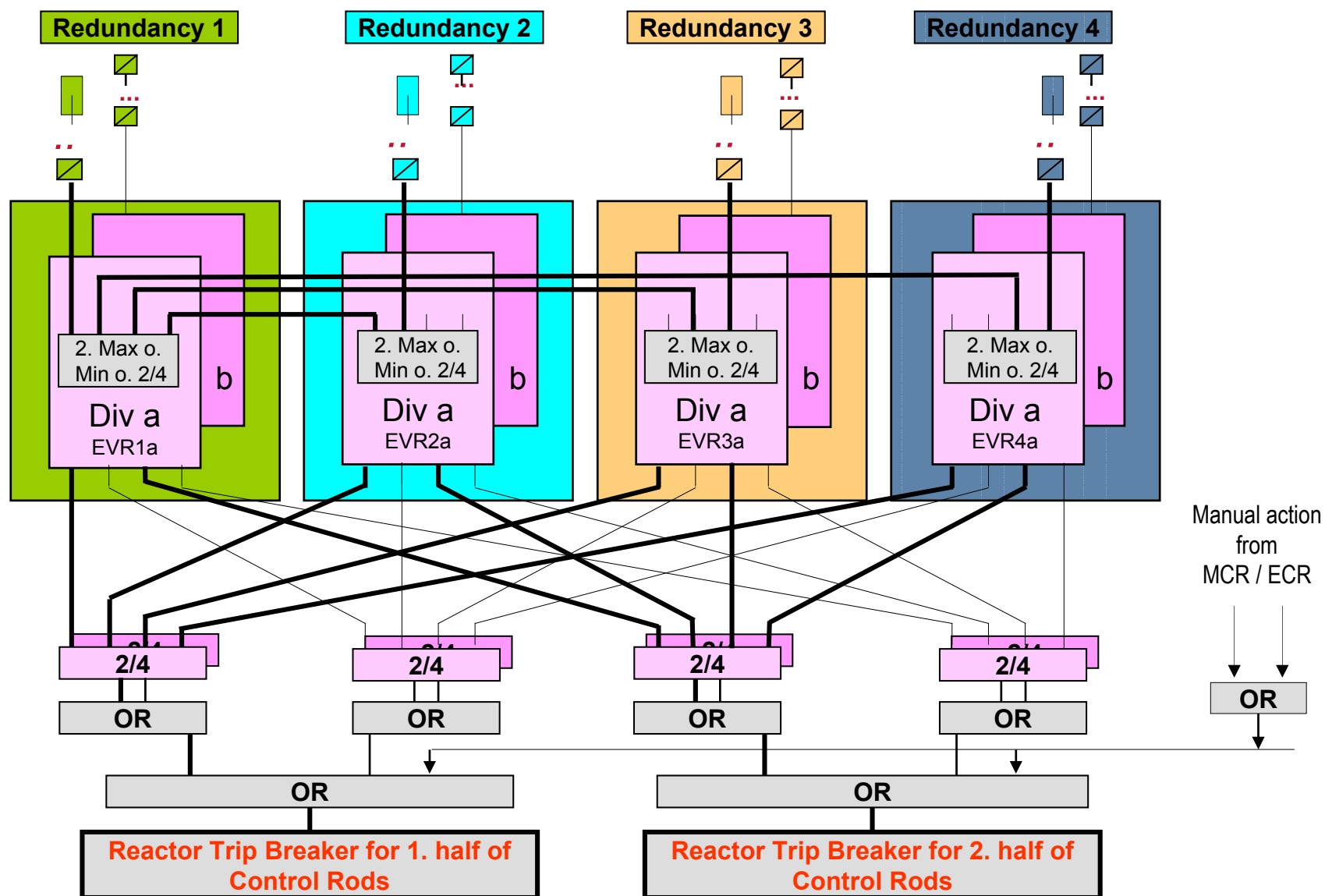
Flexible System Architecture



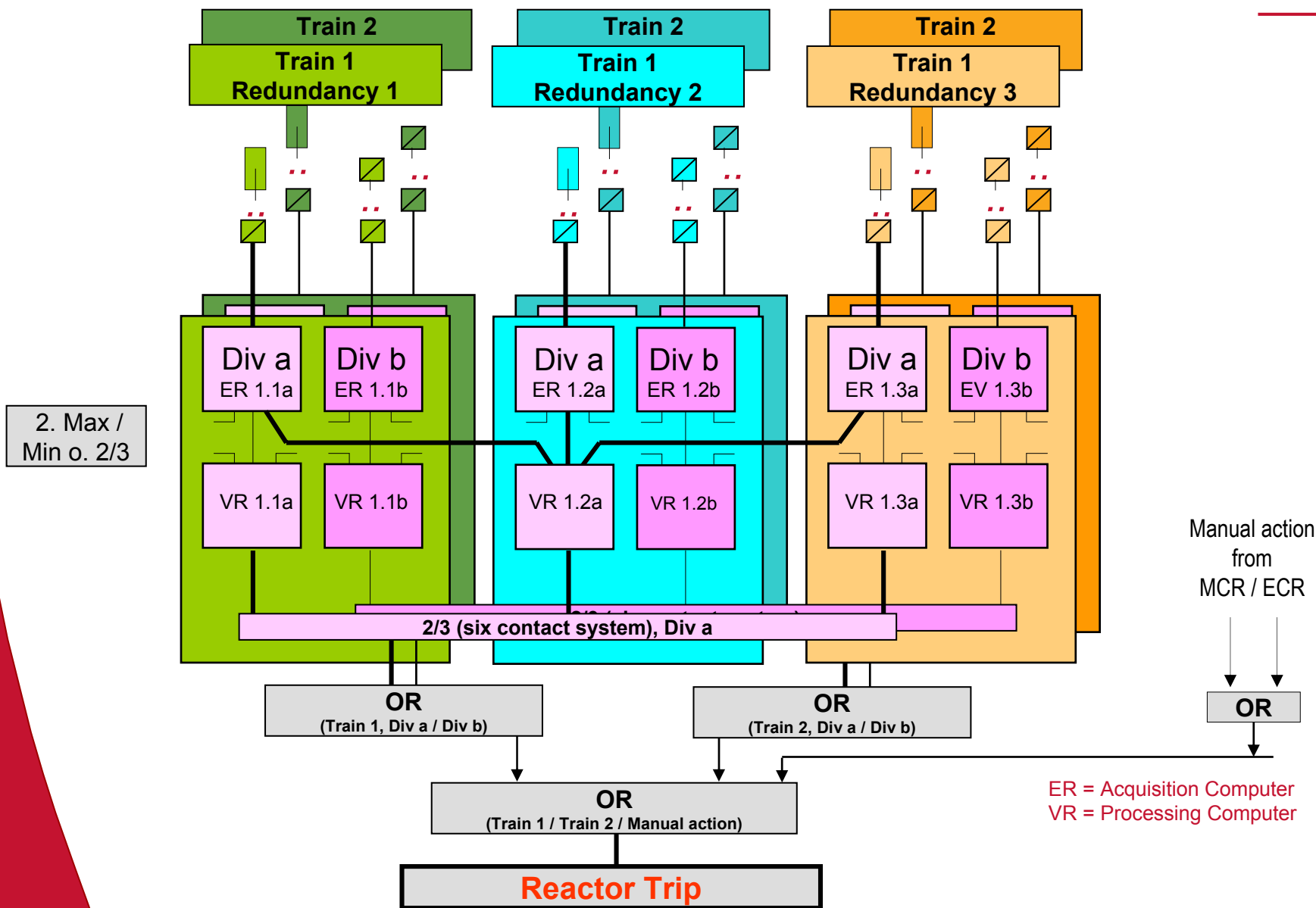
Example 1 - Reactor Trip System NPP Paks



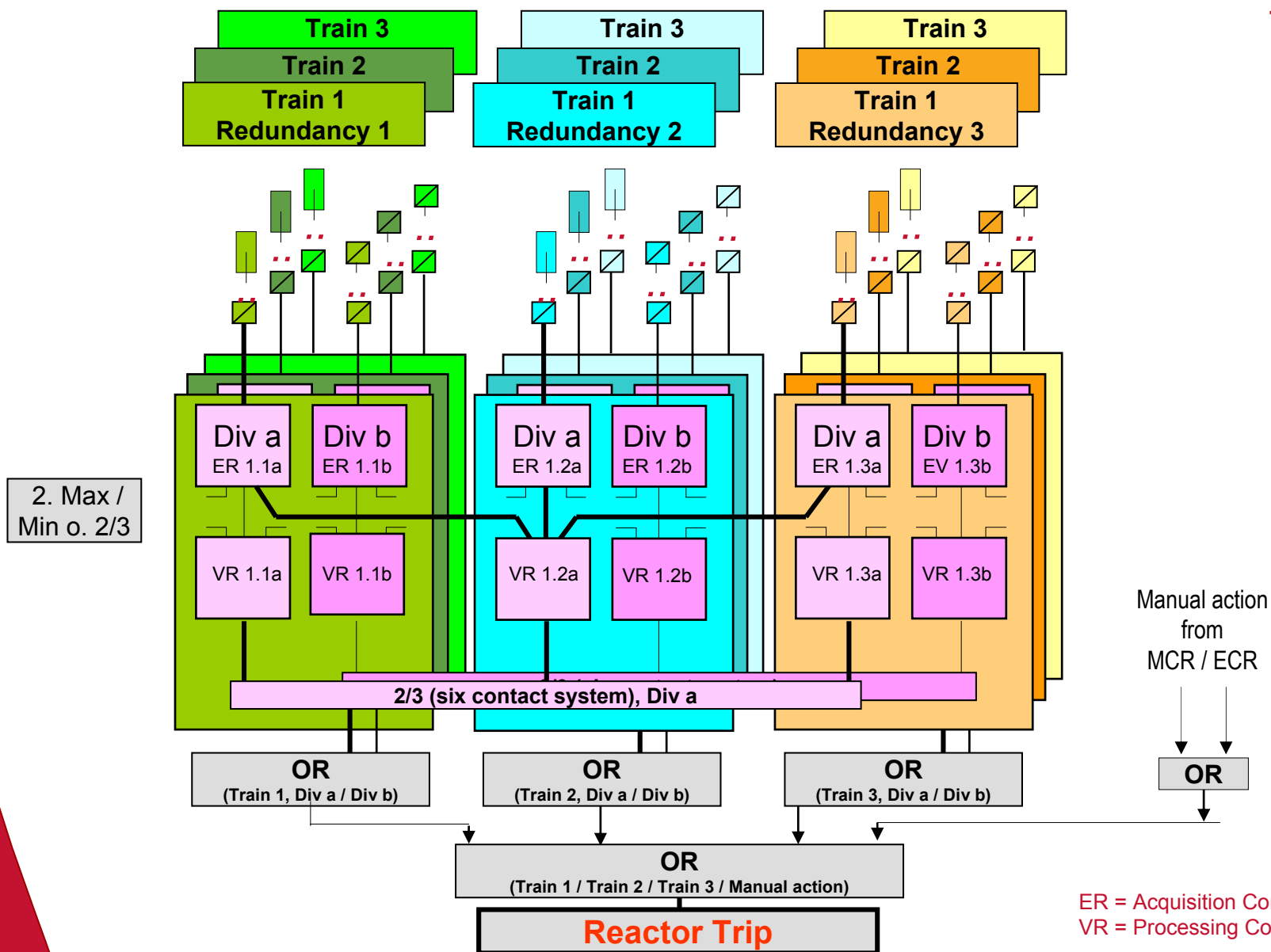
Example 2 - Reactor Trip System NPP Tianwan



Example 3 - Reactor Trip System NPP Bohunice V1

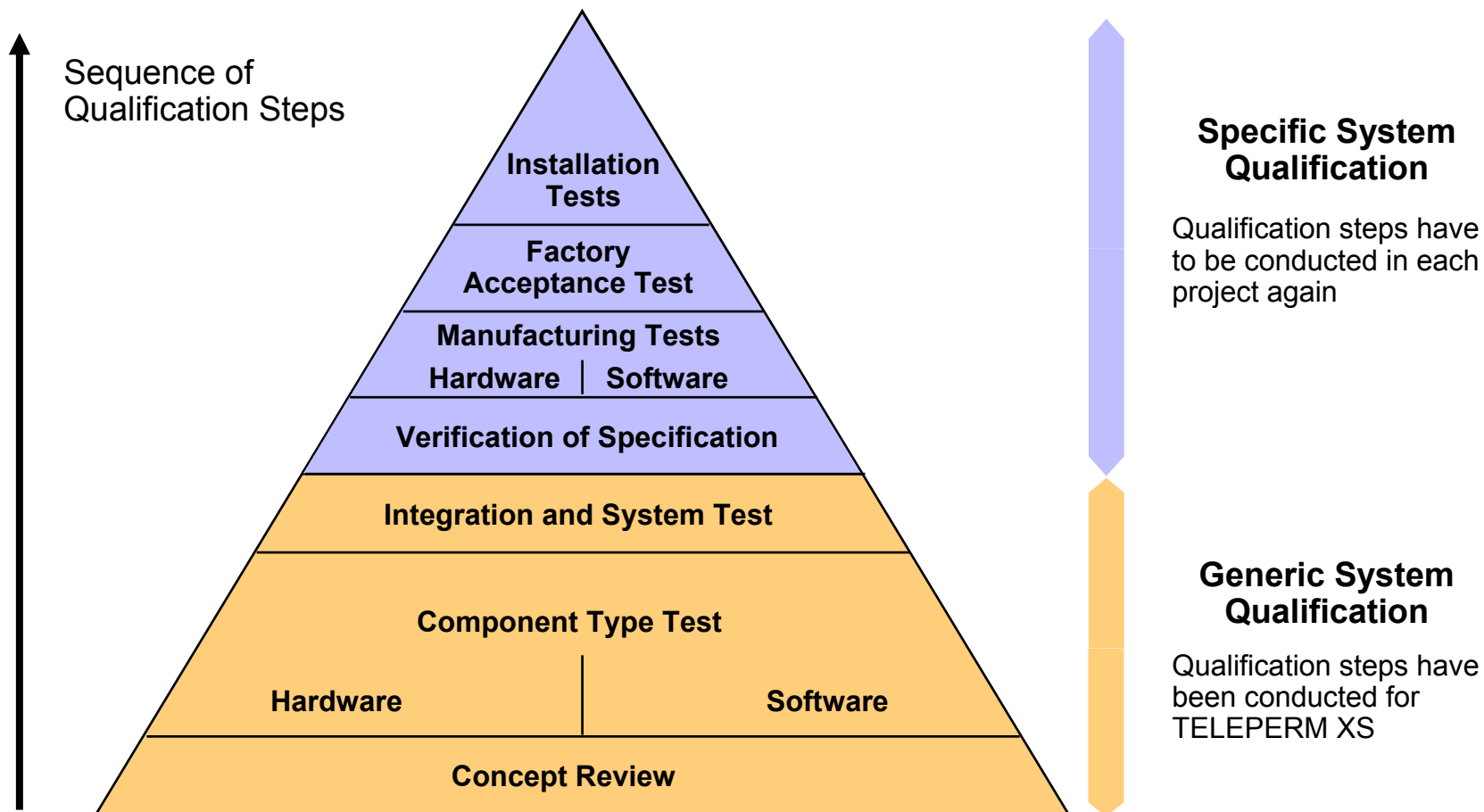


Example 4 - Reactor Trip System NPP Bohunice V2

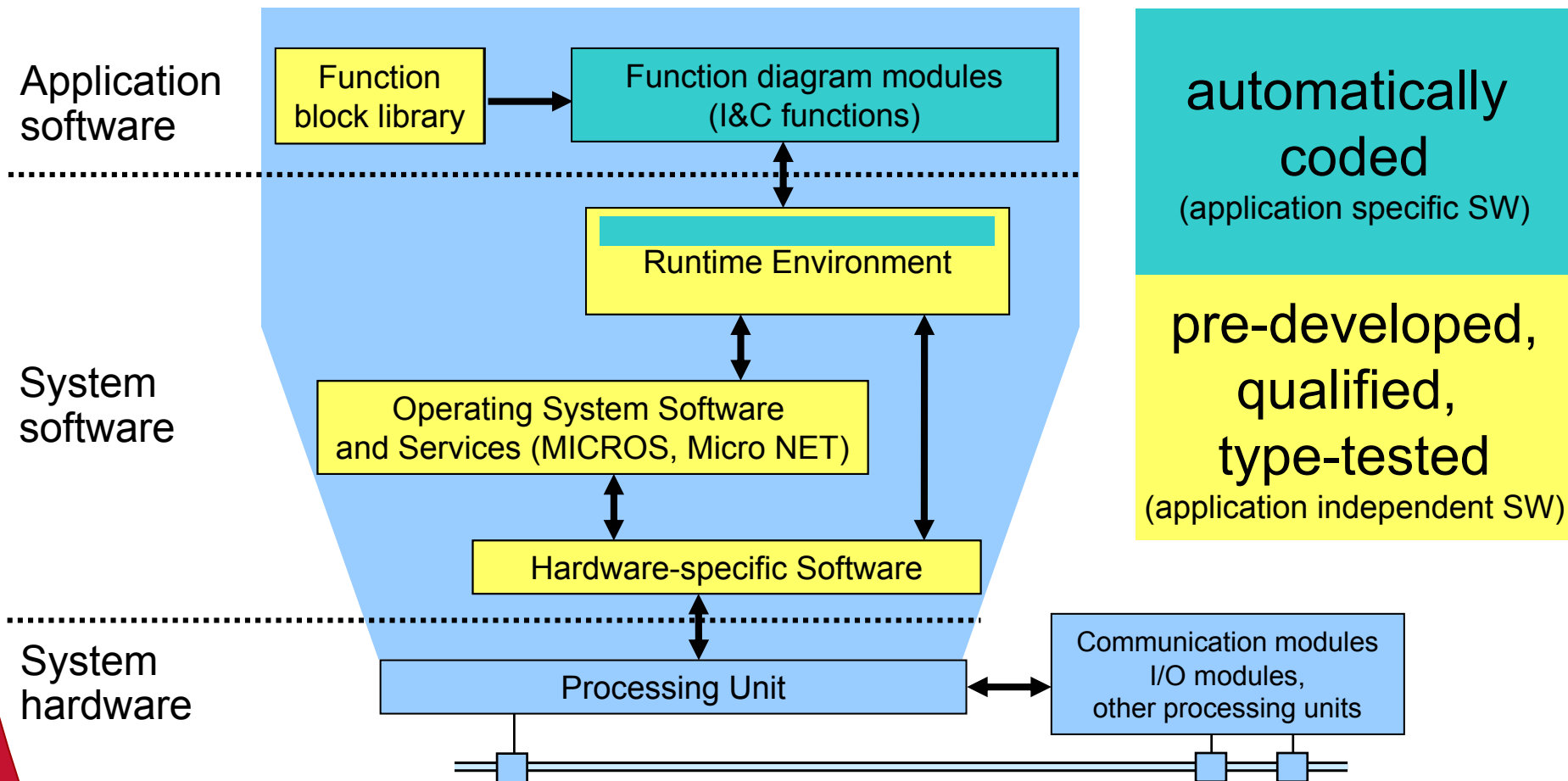


TXS Qualification Concept

Qualification Concept for TELEPERM XS



TELEPERM XS On-line Software Architecture



-  **Concept review by GRS**
-  **Software type-tests performed by ISTec and TÜV-Nord**
-  **Hardware type-tests performed by TÜV-Nord and TÜV-Rheinland**
-  **Integration and system test performed by ISTec and TÜV-Nord**

 NUREG 0800 - Standard Review Plan
Section 7 - Instrumentation and Controls

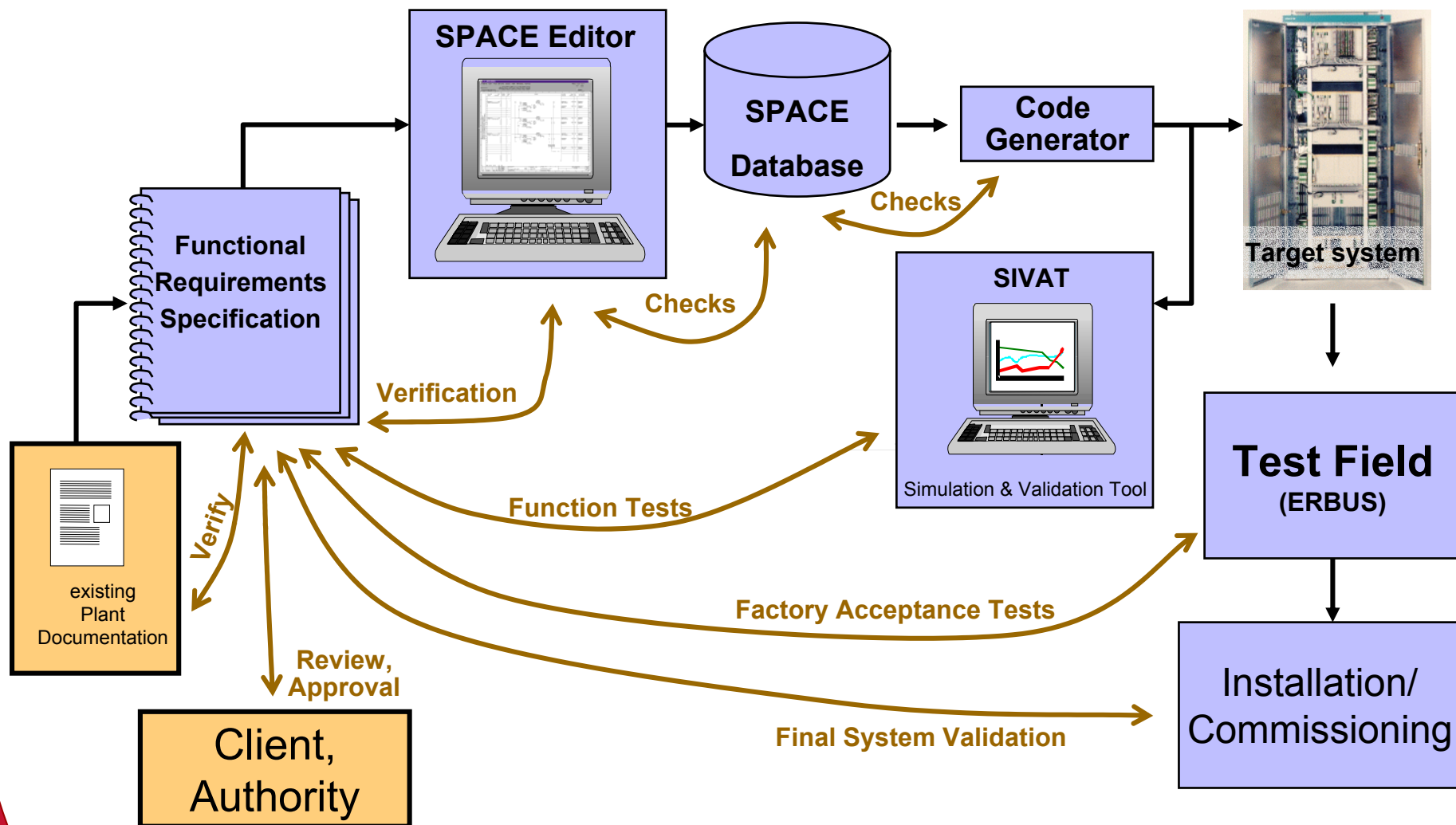
Div. Branch Technical Positions,
EPRI Topical Reports

 **NRC Conclusion:**

“Based on the information provided and the review conducted, the staff concludes that the design of the TXS system is acceptable for safety-related instrumentation and control (I&C) applications and meets the relevant regulatory requirements.”

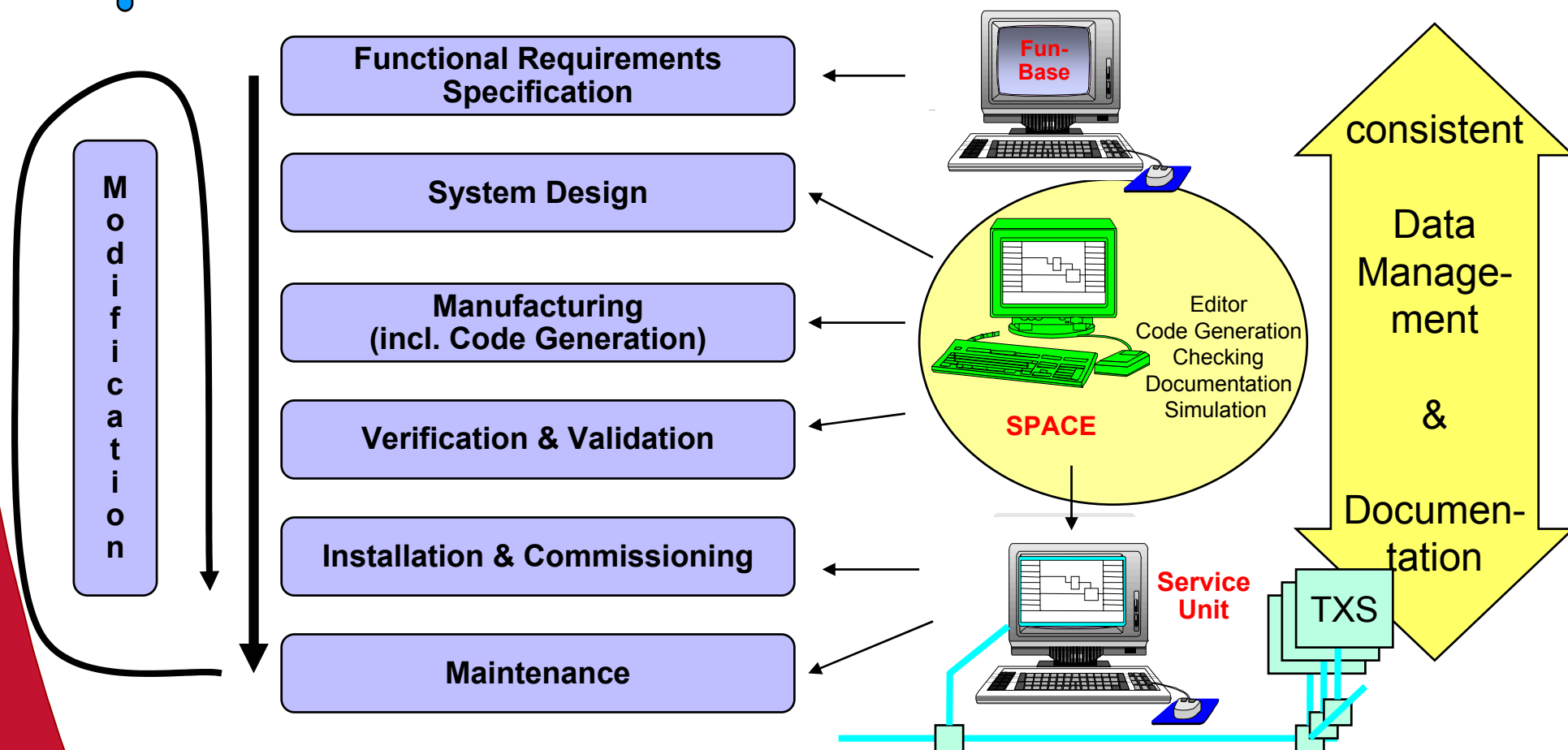
TXS Engineering Process

TELEPERM XS Engineering Process



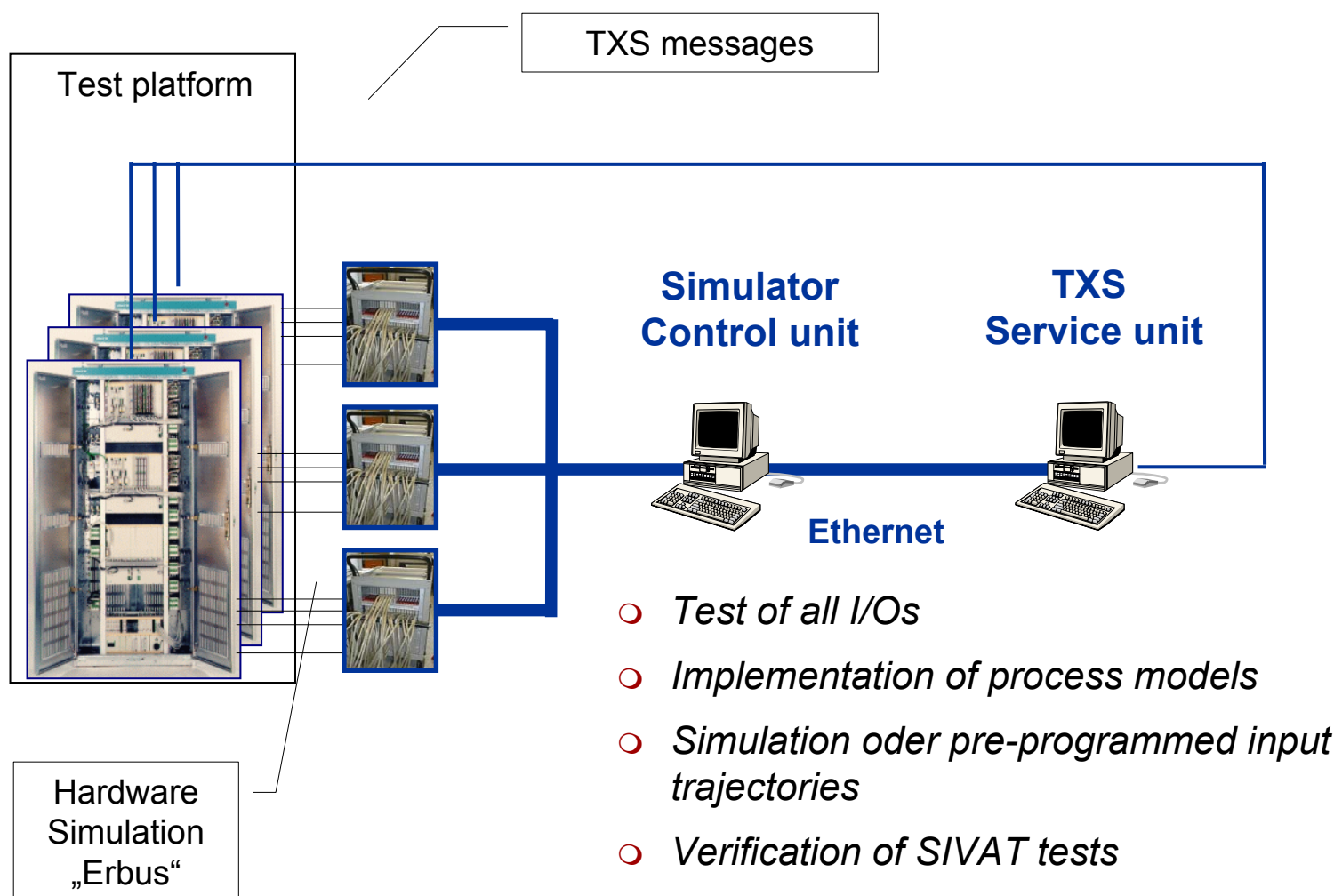
TELEPERM XS Engineering Tools

Comprehensive support of the whole I&C engineering process

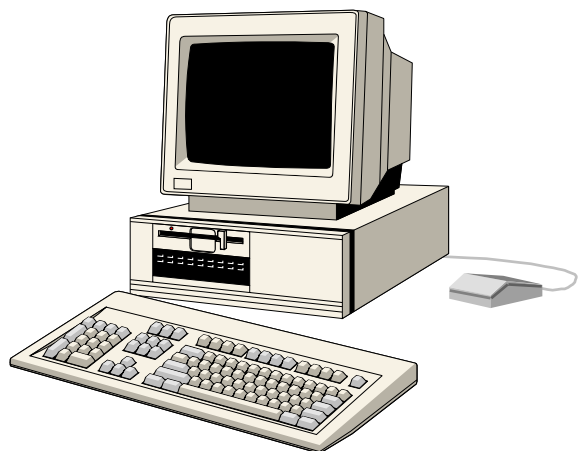


TXS Tests with SIVAT and Erbus

NPP Tianwan Interfacing to the test computer

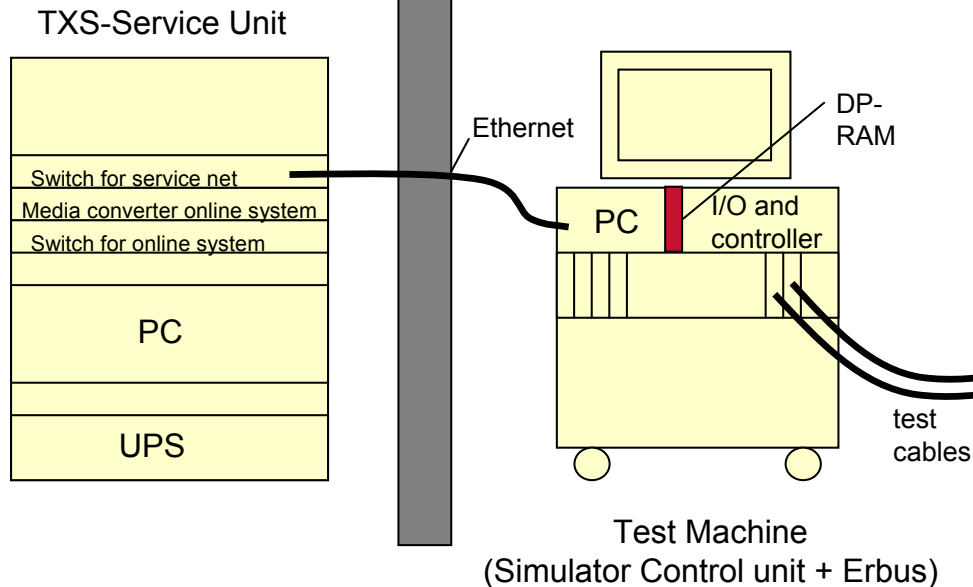
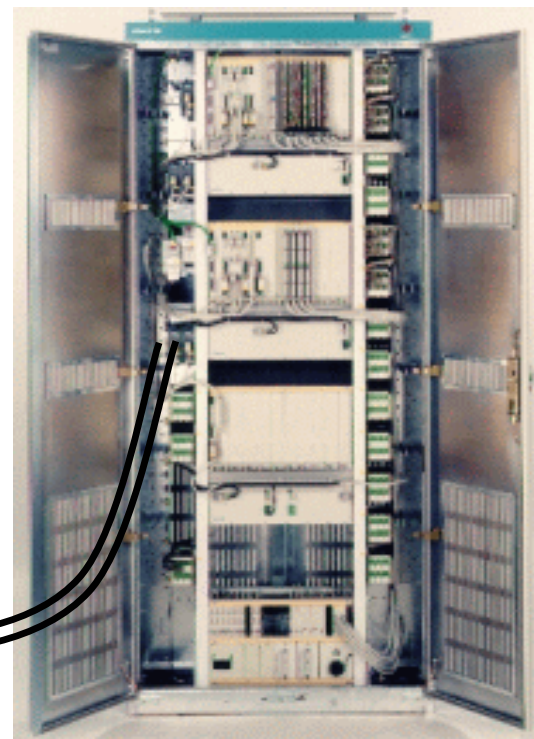


ERBUS TXS - „Test Machine“



Erbus

TXS-Cabinet



Concept for testing

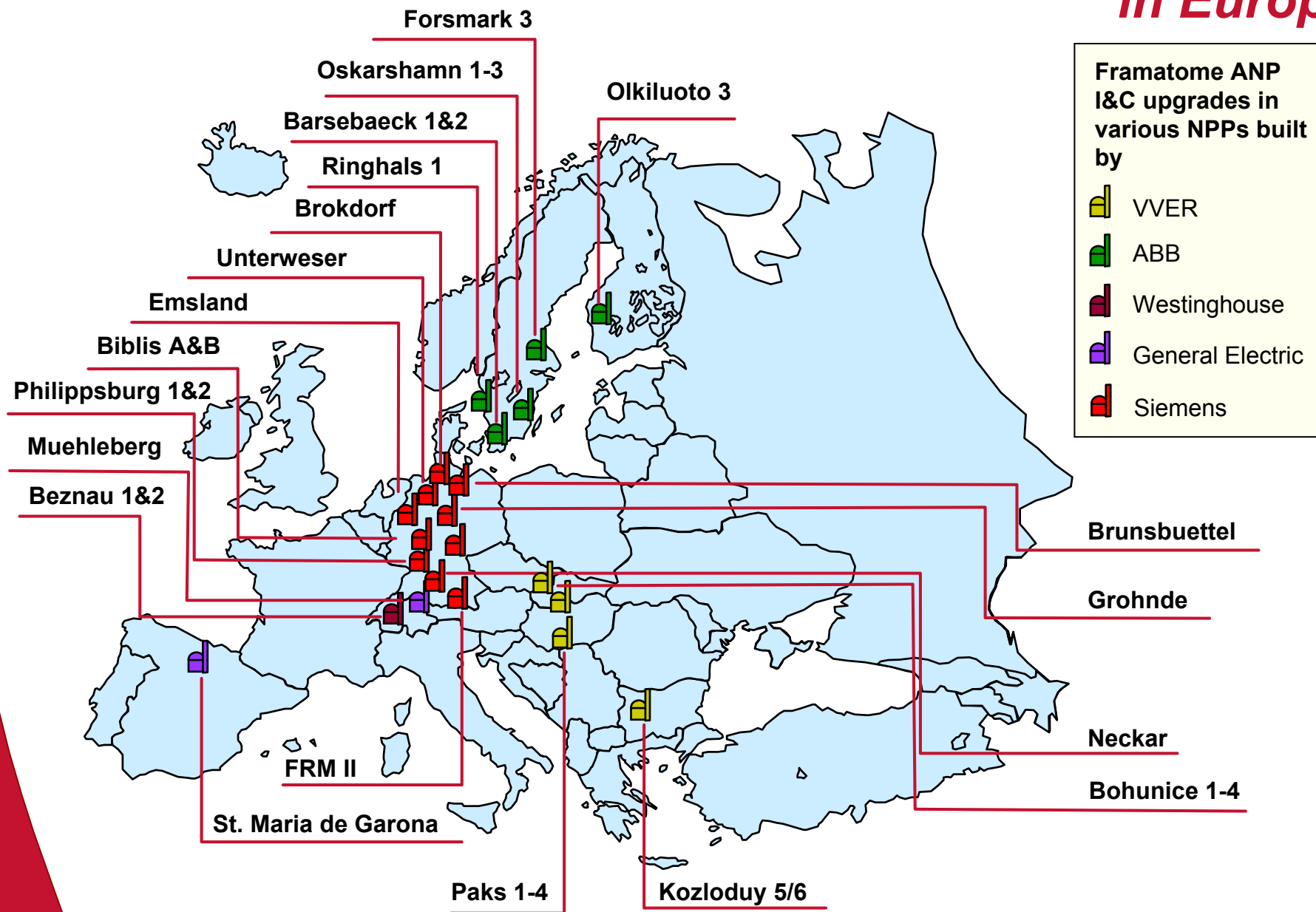
- > *Complete functional tests of TXS with simulator SIVAT in order to validate the functional requirements*
- > *Comprehensive basic test of TXS cabinets, in order to demonstrate correct manufacturing*
- > *Integral tests in the test field supported by “Erbus” concerning*
 - *Cabinet wiring and power consumption*
 - *Fault- and failure behaviour*
 - *Response time and bus/processor loads*
 - *Testability, Maintainability*
 - *Functional behaviour*
 - *Interfaces to third-party systems*

- > *Features of the modern training simulator SIVAT*
 - *All internal signals are accessible*
 - *Test of un-modified application software*
 - *Any input signal trajectories may be imposed*
 - *Simulation of hardware faults of I/O modules, processor boards, communication equipment*
 - *Easy interfacing of process models*

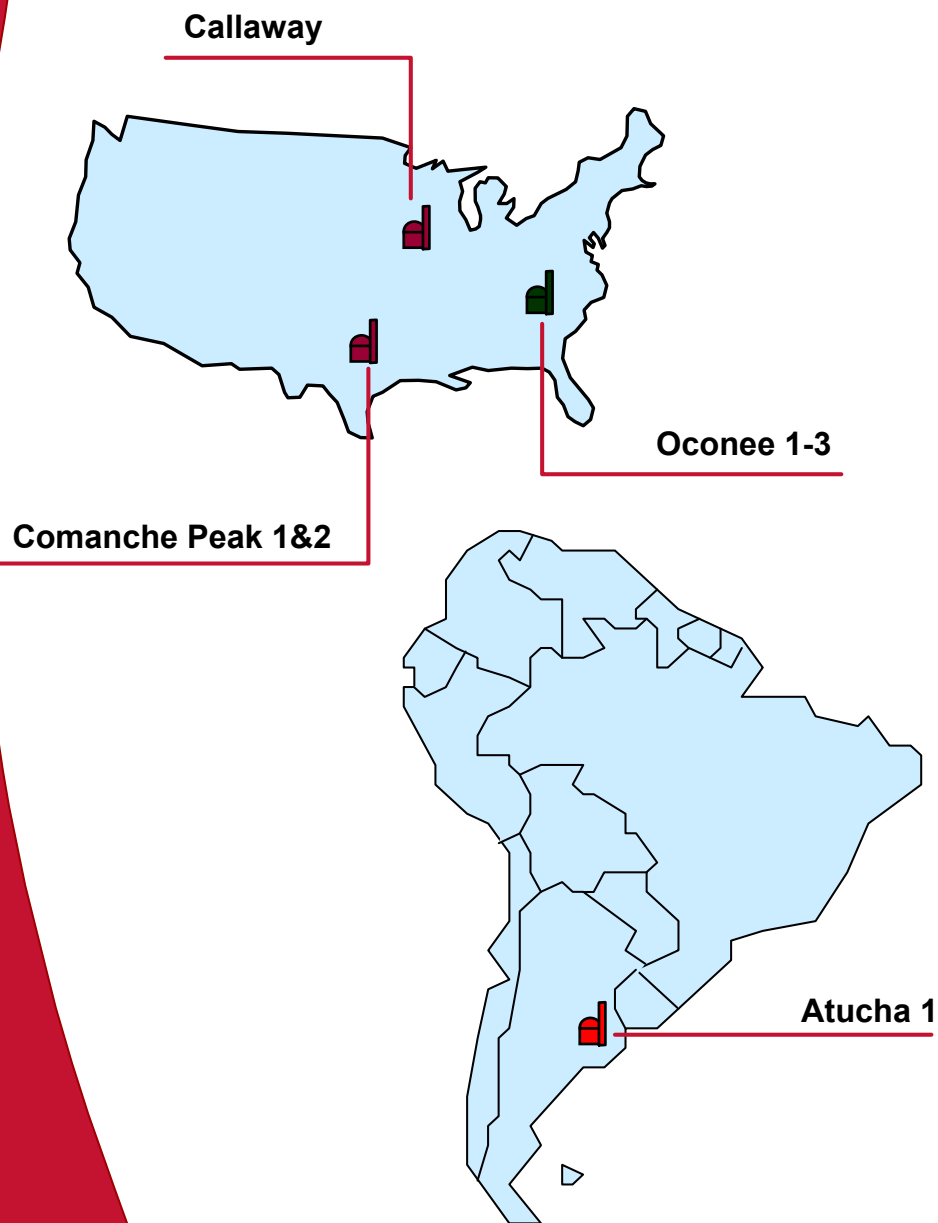
- > *The new simulator and Erbus test computer support exhaustive testing within short time*

TXS References

TELEPERM XS Applications in Europe

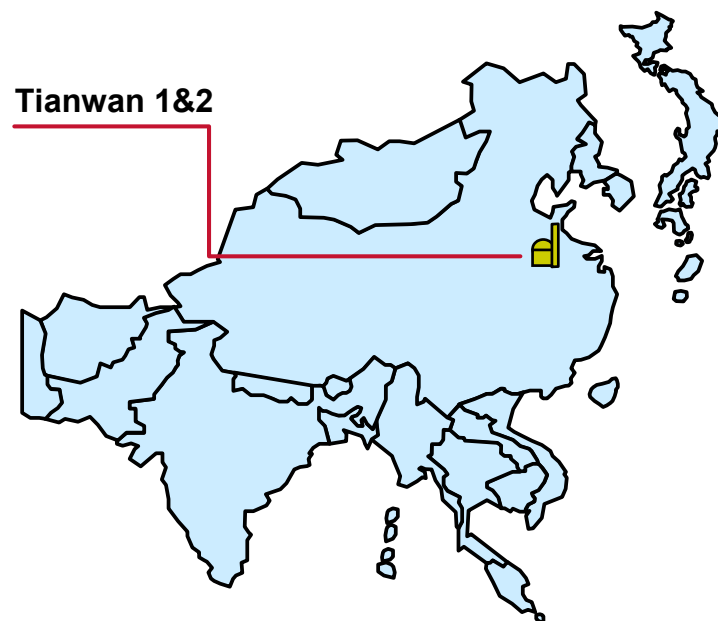


TELEPERM XS Applications in America and Asia



**Framatome ANP
TXS projects in
various NPPs built
by**

-  ASE (VVER)
-  B&W
-  Westinghouse
-  Siemens



<i>TXS operates in 23 NPP units in 8 countries</i>	<i>Application examples</i>
<i>Switzerland (3 Units)</i> <i>Germany (5 Units)</i> <i>Spain (1 Unit)</i> <i>Sweden (4 Units)</i> <i>Slovak Republic (2 Units)</i> <i>Hungary (4 Units)</i> <i>Bulgaria (2 Units)</i> <i>China (1 Unit)</i> <i>U.S.A (1 Unit)</i>	<i>Reactor trip</i> <i>Reactor Limitation</i> <i>Neutron flux measurement</i> <i>PAMS</i> <i>Subcooling margin monitoring</i> <i>Core Monitoring</i> <i>Reactor Control</i> <i>Rod Control</i> <i>Emergency Diesel sequencer</i> <i>Emergency Diesel Control</i> <i>ESFAS</i> <i>Emergency Feedwater</i> <i>Isolation of Confinement</i> <i>Turbine Protection & Control</i> <i>Plant Training Simulator</i>

Status: 2004-05-01

<i>Current TXS projects in 21 NPP units in 8 countries</i>	<i>Application examples</i>
<i>China (2)</i> <i>Germany (6)</i> <i>Sweden (1)</i> <i>Slovak Republic (2)</i> <i>Finnland (1)</i> <i>South Korea (1)</i> <i>U.S.A (6)</i> <i>Argentina (1)</i>	<i>Reactor trip</i> <i>Reactor Limitation</i> <i>Neutron flux measurement</i> <i>Coolant Overpressure Protection</i> <i>PAMS</i> <i>Subcooling margin monitoring</i> <i>Core Monitoring</i> <i>Reactor Control</i> <i>Rod Control</i> <i>Emergency Diesel sequencer</i> <i>Emergency Diesel Control</i> <i>Hydro-Turbine Governor</i> <i>Emergency Feed Water Control</i> <i>Second Heat Sink</i> <i>Turbine Protection & Control</i>

Status: 2004-05-01

TELEPERM XS

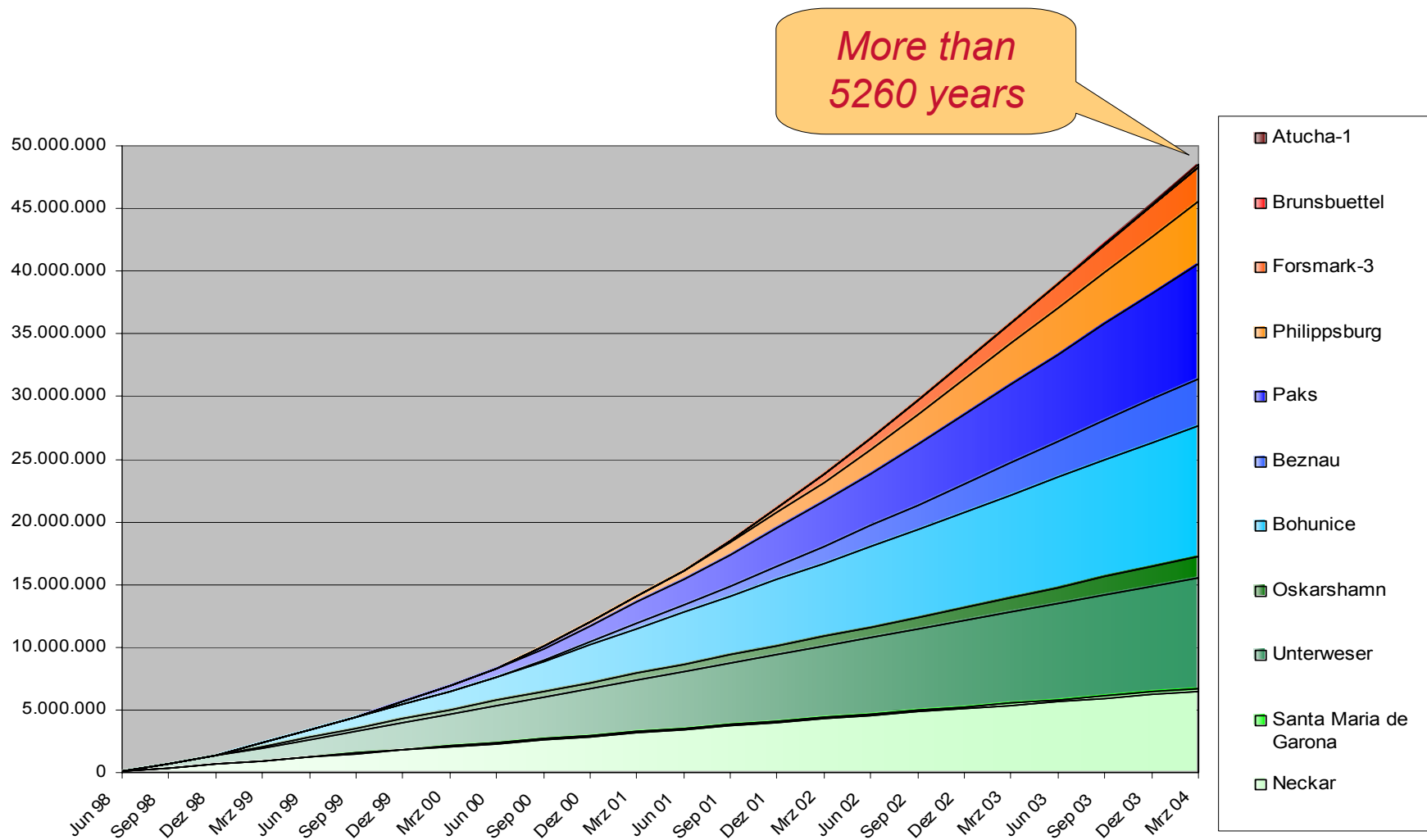
References

Project	Number of Processing Modules per Unit
<i>Tianwan 1&2 (China)</i>	250
<i>Unterweser (Germany)</i>	177
<i>Bohunice V2, 3&4 (Slovak Rep.)</i>	151
<i>Bohunice V1, 1&2 (Slovak Rep.)</i>	132
<i>Beznau 1&2 (Switzerland, Pressure, Ergres)</i>	130 + 4
<i>Forsmark 3 (Sweden, MODS)</i>	126
<i>Phillipsburg 1 (Germany, Redundance)</i>	121
<i>Neckar 1 (Germany)</i>	87
<i>Paks 1-4 (Hungary)</i>	62
<i>Kozloduy 5&6 (Bulgaria, Diesel Control, COP)</i>	12 +12
<i>Beznau (Switzerland, Erges)</i>	8
<i>Garona (Spain)</i>	4

Status: 2004-05-01

TXS Operational Experience

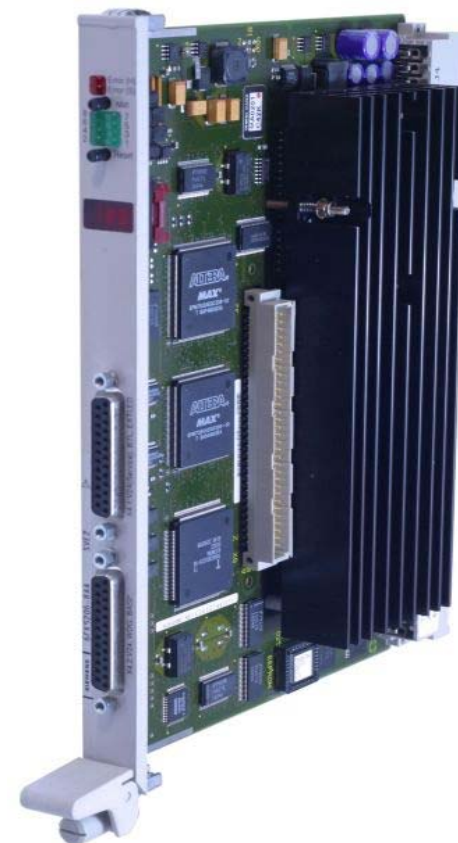
TELEPERM XS Processor Moduls Operating Hours



Status: 2004-04-01

TELEPERM XS Excellent Reliability

<i>Component</i>	<i>Processor Modules</i>	<i>Average of all Platform Modules</i>
<i>Components in Operation</i>	1 439	14 124
<i>Operating Hours</i>	48 500 304	450 265 560



Status: 2004-04-01

TELEPERM XS Processor Modules Failure Rate: $2,89 * 10^{-7}$ [1/h]
An excellent result compared to implemented analog systems

TXS Outlook

Continuous improvement of our Safety I&C system

TELEPERM XS

Project FUTIS

**Future
I&C
for
Safety
Applications**

*Common product strategy for German,
French and US-business lines*

*Focus on international market requirements
and long term support*

Further development of TXS TELEPERM XS Hardware

New Generation of Communication modules

- 12 MBit/s TXS-Profibus L2 (now 1,5 MBit/s)
- 100 MBit/s TXS-Ethernet H1 (now 10 MBit/s)

New Generation of I / O Modules

Downwards compatible

- + Wire break control
- + Further improvement of self diagnostic features
- + Time stamp of I/O signals

Improved support of decentralized solutions

- decentralized I/O modules
- decentralized compact processing units

Framatome ANP has developed
a long term strategy
for the support and further development
of TELEPERM XS

Thank you for your attention